Mitosis Video

1. Most plant growth takes place just behind the _____ _____.
2. When viewed under a microscope, the nuclei of some cells look unusual. Individual _____ look different from one another.
3. In the growing region, the nuclei contain threadlike _____.
4. Mitosis is a process where a single cell _____ to form two cells.
5. The two threads making up the chromosomes are called _____.
6. The chromosomes move towards the _____ of the cell.
7. The chromatids of every chromosome _____ from one another and move to _____ _____ of the cell.
8. The chromatids are now new _____.
9. A non-dividing cell is in the period of _____.
10. During prophase, each chromosome becomes _____ and each consists of two _____.
11. During _____ the chromosomes meet at the centre of the cell.
12. Anaphase occurs when the chromosomes _____.
13. The final period of mitosis is called _____.
14. Before mitosis begins, each chromosome has replicated to give two chromatids. The chromatids are held together at the _____.
15. The centromeres become attached to very fine threads called _____ _____.
16. During anaphase in an animal cell, the _____ _____ constricts across the centre dividing the _____ into two halves.
17. The _____ consists of two cylinder shaped structures at right angles to each other.
18. The centrioles move apart and _____ _____ appear between them. Together they form the spindle.

19. Fibres that radiate out from each centriole form a structure called an _____.
20. The centre of the spindle is called the _____.
21. The chromatids are pulled to opposite poles by _____ of the spindle fibres.

Meiosis Video

1. Human cells have 46 chromosomes.
2. During meiosis similar chromosomes come together in pairs.
3. The members of each pair move to opposite ends of the cell.
4. After two cells are formed, each cell divides again but there is no pairing of chromosomes. The chromatids separate and move to opposite ends of the spindle.
5. At the end of the second meiotic division there are four daughter cells each with half the number of chromosomes of the original cell.
6. The matching pairs are homologous pairs already made of two chromatids.
7. Chiasmata are cross-over points where chromosome material is exchanged.
8. The formation of the spindle during the second meiotic division is often at right angles to the first.
9. One cell with a pair of chromosomes divides to form 4 daughter cells, each with a single chromosome.
10. Before meiosis, each cell is diploid represented by 2n.
11. After meiosis, 4 haploid cells each with n chromosomes result.
12. In females the cell does not divide into equal halves. Three small polar bodies are formed and eventually break down.
13. At fertilisation the egg(n) and the sperm(n) combine to form a zygote(2n).

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